Graph Theory

What is a graph?

A graph G is a tuple, i.e., $G := (V, E)$,

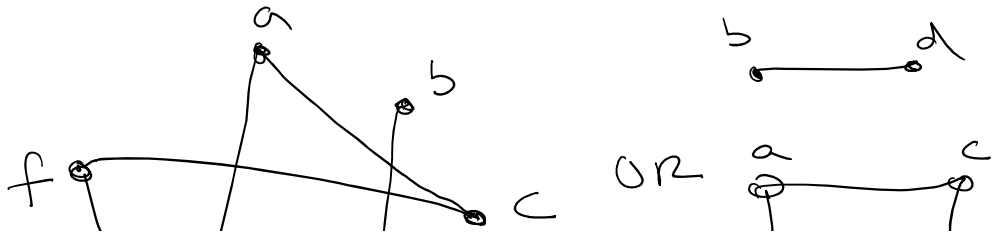
where V (or $V(G)$) is a set of elements called the vertex set (or node set) (each element in this set is called a vertex (or node)) and E (or $E(G)$) is a collection of 2-sized subsets of V and is called the edge set (each element in E is called an edge).

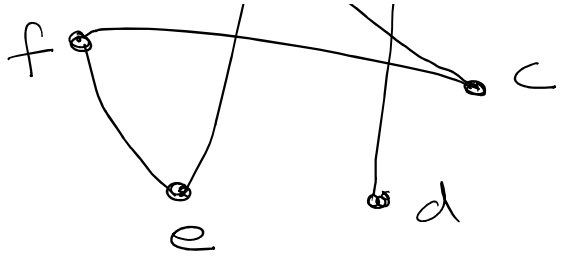
Example

$G = (V, E)$, where

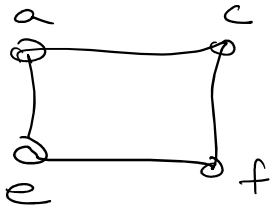
$$V = \{a, b, c, d, e, f\}$$

$$E = \{\{a, c\}, \{a, e\}, \{b, d\}, \{e, f\}, \{c, f\}\}$$





OR



A more general notion

Hypergraph → is a tuple

$H = (V, E)$

V → vertex set (a collection of elements)

~~2-sized~~

E → a collection of subsets of V.

(a subset in E is called a hyperedge)

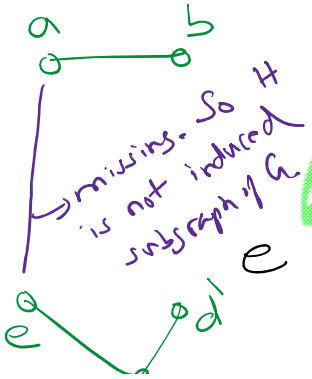
Example $H = (V, E)$

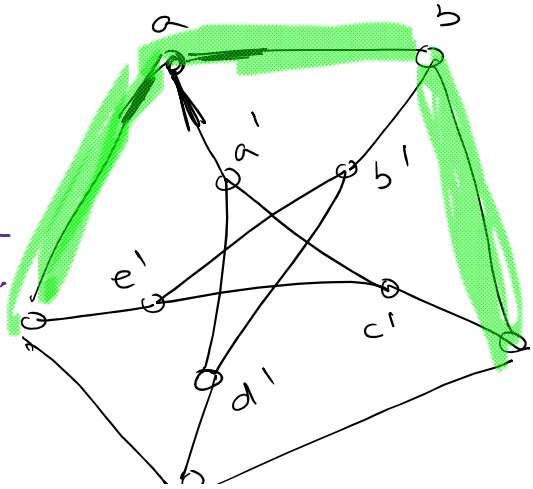
$V = \{a, b, c, d, e, f\}$

$E = \{\{a, b, c, d\}, \{c, e, f\}, \{a, e, f\}, \{b, d\}\}$

Some definitions on graphs

$H = (V', E')$



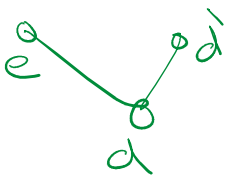


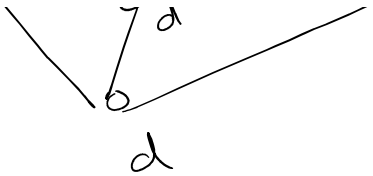
$G = (V, E)$

$V = \{a, b, c, d, e, a', b', c', d', e'\}$

$E = \{\{a, b\}, \{b, c\}, \{c, d\}, \{d, e\}, \{e, a\},$

OR [illegible]





$\{\{a,e\},\{e,c\},$
OR
(a,b) $\{a,a'\},\{b,b'\}$
OR
ab $\{e,e'\}$
$\{a',c'\},\{c',e'\}$
$\{e',b'\},\{b',d'\}$
$\{d',a'\}\}$

$|V| = 10$

$|E| = 15$

a is a neighbour of b

$N_G(a) = \{b, a', e\}$

Notion

(i) degree of a vertex $\longrightarrow$ no. of edges it is present in.

$$\deg_G(a) = 3, \quad \deg_G(b) = 3$$

(ii) maximum degree of G, denoted by $\Delta(G)$, is

$$\Delta(G) := \max\{\deg_G(v) : v \in V(G)\}$$

above example, $\Delta(G) = 3$.

(iii) minimum degree of G, denoted by

$$\delta(G) := \min\{\deg_G(v) : v \in V(G)\}$$

above example, $\delta(G) = 3$

(iv) given two vertices u and v, we say u is adjacent to v

if $\{u, v\} \in E(G)$.

u is a neighbor of v in G if $\{u, v\} \in E(G)$.

(v) Neighborhood of a vertex v in G is denoted $N_G(v)$ and is defined as $N_G(v) = \{u : \{u, v\} \in E(G)\}$

(vi) $H = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

(vii) a vertex v is incident with an edge e if $v \in e$.
↳ 2-sized subset of V.

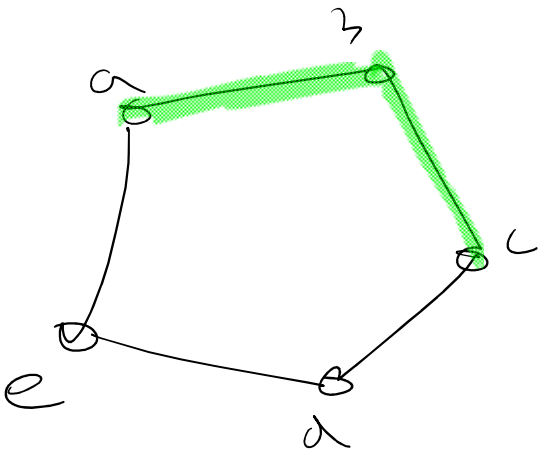
(ix) $H = (V', E')$ is an induced subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' = \{(u, v) \in E(G) : u, v \in V'\}$

(x) $H = (V', E')$ is a spanning subgraph of $G = (V, E)$ if H is a subgraph of G and $V' = V$.

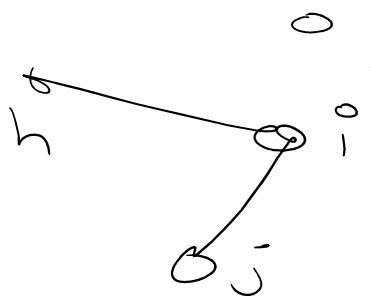
↳ spanning

(xi) Average degree of a graph G, denoted $d_{avg}(G)$ is

$$d_{avg}(G) := \frac{\sum_{v \in V(G)} deg_G(v)}{|V|}$$



for $H = (V, E)$



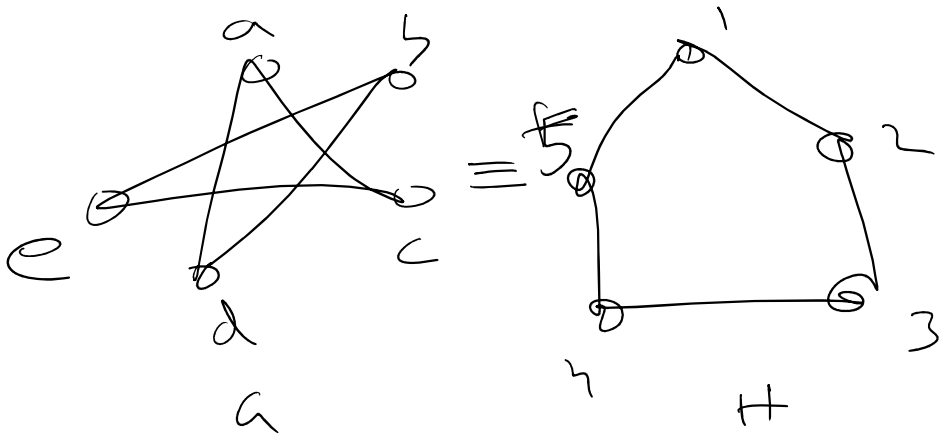


$V(H) = \{a, b, \ldots, l\}$

$E(H) = \{\{a,e\}, \{a,b\}, \ldots, \{k,l\}\}$

* sum of degrees $= 2 \cdot |E(G)|$.
Therefore no. of odd degree vertices is even.





G is isomorphic to H.

$f : V(G) \to V(G)$ as

$f(a) = 1$, $f(c) = 2$, $f(e) = 3$, $f(b) = 4$,

$f(d) = 5$

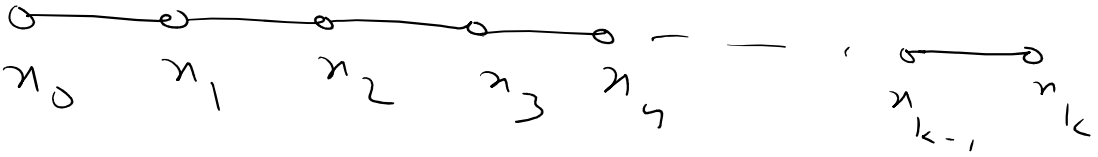
* A <u>path</u> is a non-empty graph

$P = (V, E)$, where

$V = \{x_0, x_1, x_2, \ldots, x_k\}$

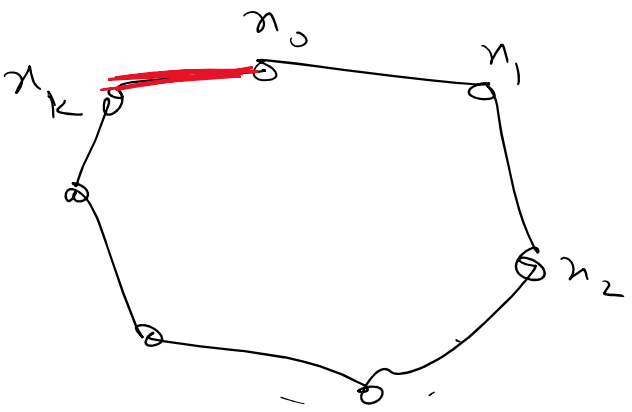
$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \{x_2, x_3\}, \ldots\}$

$$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \{x_2, x_3\}, \ldots \ldots \{x_{k-1}, x_k\}\}$$



length of a path is the no. of edges in it.

* Cycle



Let $P = (V, E)$ be a path, where $V = \{x_0, x_1, \ldots, x_k\}$

$$E = \{\{x_0, x_1\}, \{x_1, x_2\}, \ldots, \{x_{k-1}, x_k\}\}$$

Then, $P + \{x_k, x_0\}$ is a cycle

Lemma Every graph has (i) a path of length $\delta(G)$, and

starting at a vertex with min degree

start with min degree

(i) a path of length $\delta(G)$, and

(ii) a cycle of length $\delta(G)+1$, $(\delta(G) \geq 2)$

Proof.

Proving (i) Consider a graph G with min degree $\delta(G)$.

By mathematical induction on no. of vertices.

Base case: $|V(G)| = |V| = 1$.

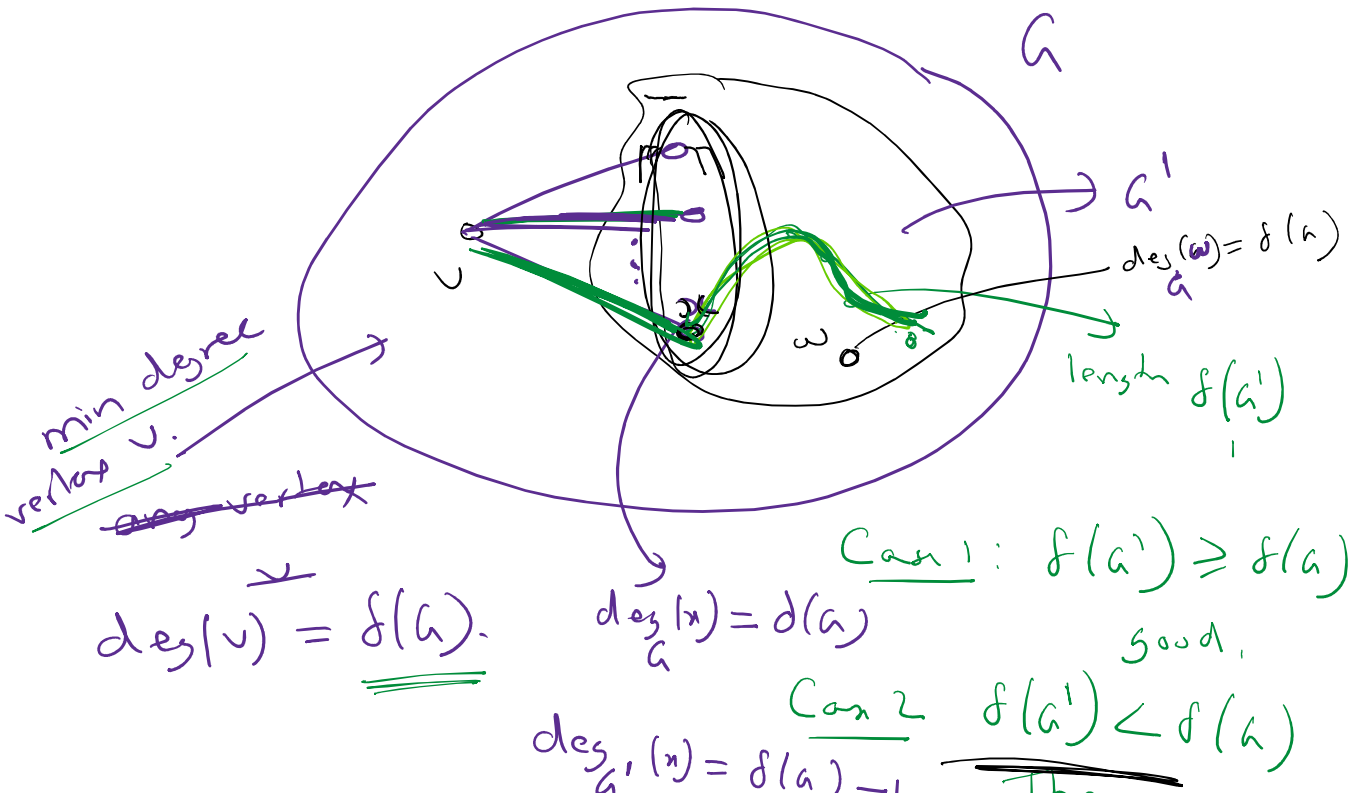
$\delta(G) = 0$.

0-length path

Induction step.

Assume the stmt is TRUE for all graphs on at most n vertices.

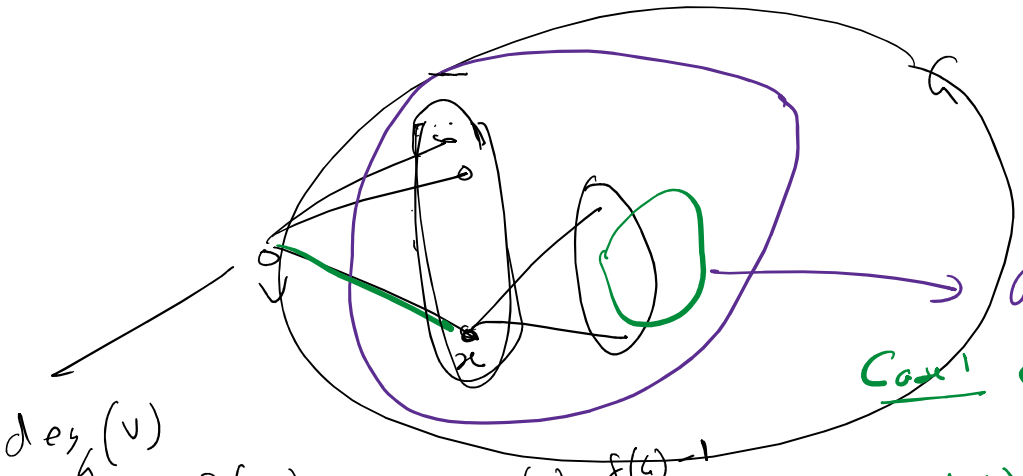
Suppose G has $n+1$ vertices.



$\deg_{G'}(x) = \delta(G) - 1$ Case 2 $\delta(G') < \delta(G)$

Then,

$\delta(G') = \delta(G) - 1$



$\deg_G(v) = \delta(G)$ $\deg_{G'}(x) = \delta(G) - 1$

Case 1 $\delta(G') \geq \delta(G)$.

Good

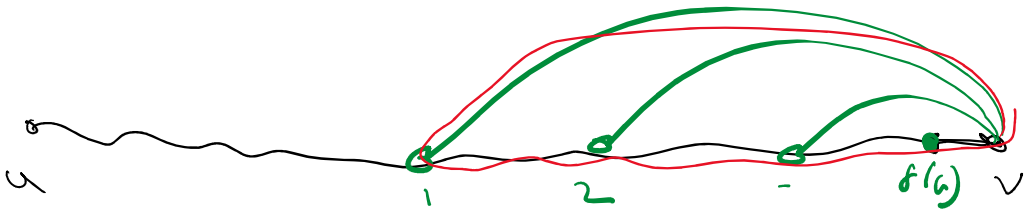
Case 2 $\delta(G') = \delta(G) - 1$

↳ How to go forward ??

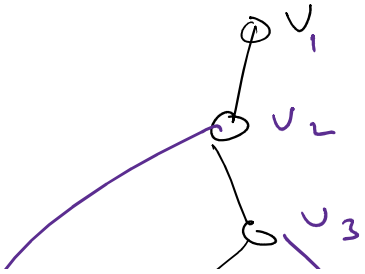
Alternate approach

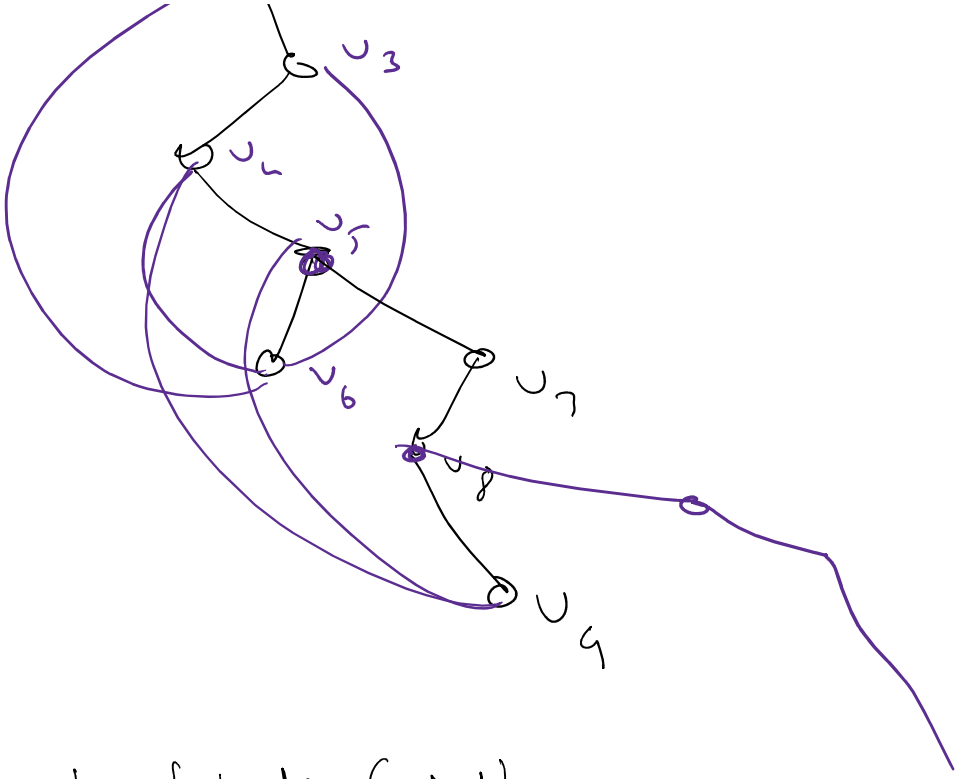
Proving (i) and (ii)

Take the longest path in G.



$\deg_G(v) \geq \delta(G)$





Lecture by Subrata (Wed)

Thursday

Recap:

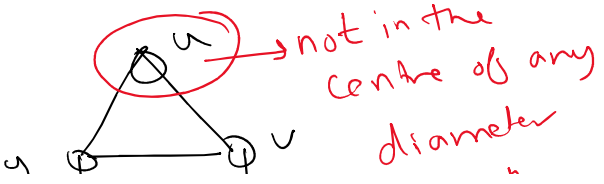
radius, diameter, central vertex, girth, circumference, distance.

Stmt. Given a graph G, of G.

(i) Every central vertex is in the centre of a diameter path

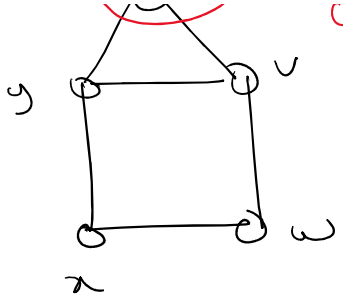
(ii) The centre of a diameter path is a central vertex of G.

Statement (i) is wrong



Statement (ii) is wrong



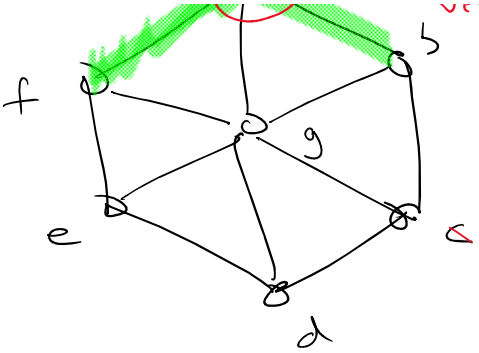


diameter path.

radius = 2

diameter = 2

every vertex is a central vertex.



radius = 1

diameter = 2

g is a central vertex of G.

Proposition. $rad(G) \leq diam(G) \leq 2 \cdot rad(G)$.

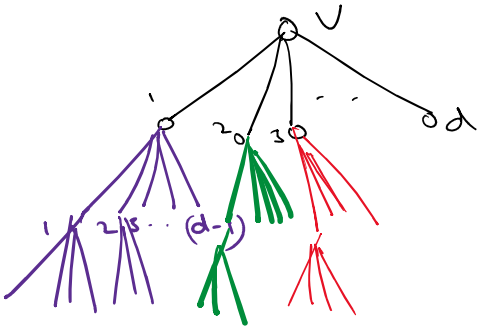
Proposition $radius(G) \leq k$

max deg of $G \leq d$, where $d \geq 3$

Show that the graph G has fewer than

$$\frac{d}{(d-2)}(d-1)^{k}$$ vertices.

Proof: Let v be a central vertex of G.



Level 0 → 1

Level 1 → d

Level 2 → $d(d-1)$

Level 3 → $d(d-1)^2$

Level k → $d(d-1)^{k-1}$

Level $k \to d(d-1)^{k-1}$

$$\#\text{ vertices} \le 1 + d + d(d-1) + \cdots + d(d-1)^{k-1}$$

$$\le 1 + d\left(1 + (d-1) + \cdots + (d-1)^{k-1}\right)$$

$$= 1 + d\left(\frac{(d-1)^k - 1}{d-2}\right)$$

$$= \frac{d(d-1)^k}{d-2} - \frac{d}{d-2} + 1$$

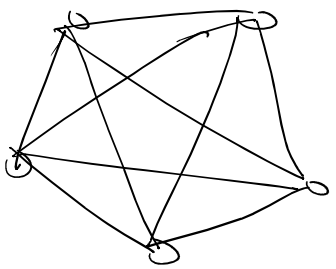
$$< \frac{d(d-1)^k}{d-2}$$

//

Some commonly seen graphs

(i) Complete graph on n vertices, denoted K_n, and it has all the $\binom{n}{2}$ edges.

K_5



Some definitions

* Independent set :

* Independent set: Given a graph G, $S \subseteq V(G)$ is called an independent set, if no two vertices in S have an edge between them.

Independence no. of a graph G, denoted $\alpha(G)$, is the size/cardinality of a largest independent set in G.

* Clique: A set $S \subseteq V(G)$ is a clique if every two vertices in S have an edge between them in G.

Size/cardinality of a largest clique in G is called clique number of G. It is denoted $\omega(G)$.

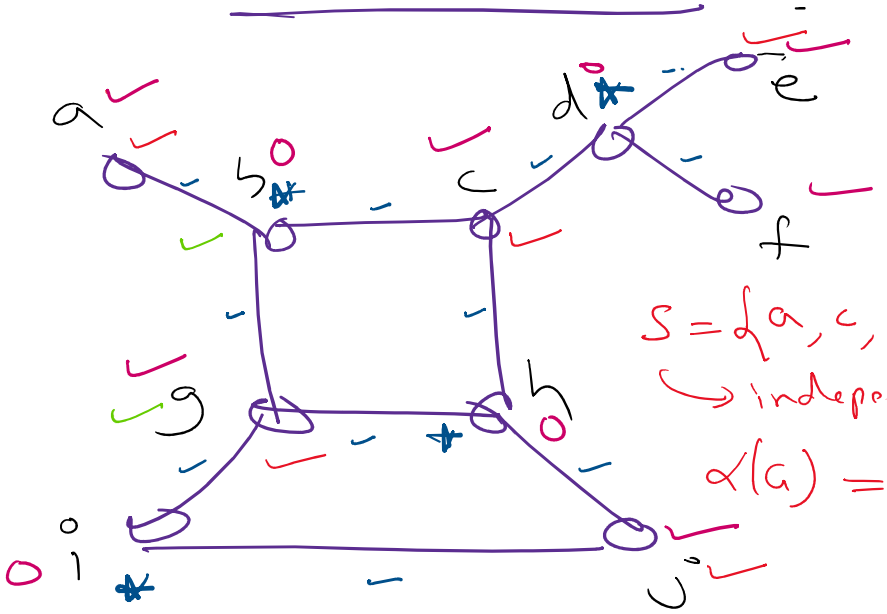
* Vertex cover: A set $S \subseteq V(G)$ is a vertex cover of G if every

edge in G has an endpoint in S.

A smallest cardinality vertex cover of G is called a minimum vertex cover

* Dominating set : A set $S \subseteq V(G)$ is called a dominating set of G if every vertex in G is either in S or has a neighbor in S.

Example to explain the notion



$S = \{a, c, e, g, j\}$ → independent set

$\alpha(G) = 6$

$S' = \{b, g\}$ is a clique.

$S'' = \{b, d, h, i\}$ is a vertex cover

$S'' = \{b, d, h, i\}$ is a vertex cover.

Is it the smallest one?

$S''' = \{b, d, h, i\}$ is a dominating set.

Is this a smallest dominating set?

Coming back to examples of common graphs

$\alpha(G)$ → max indep. set size

$\beta(G)$ → min vertex cover size

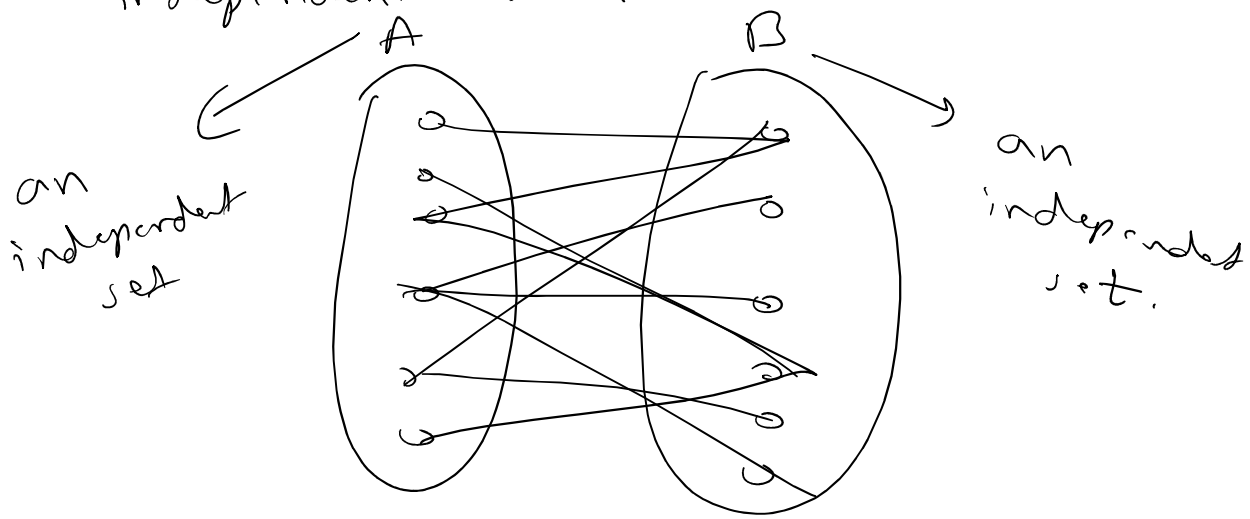
$\gamma(G)$ → largest dominating set size.

(i) Complete graph on n vertices

$\alpha(G) = 1$, $\beta(G) = n-1$, $\gamma(G) = 1$

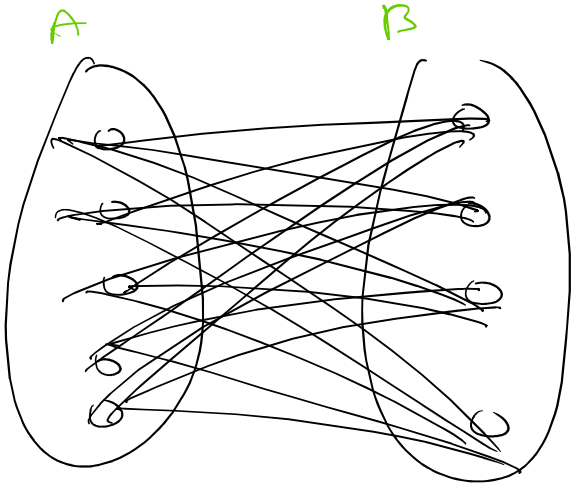
(ii) Bipartite graph:

A graph whose vertices can be partitioned into two independent sets.



(iii) Complete bipartite graph

A B $\alpha(G) = \max(|A|, |B|)$

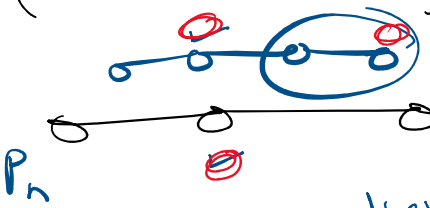


$\alpha(G) = \max(|A|, |B|)$

$\beta(G) = \min(|A|, |B|)$

$\gamma(G) = 2$

(iv) Path graph on n vertices



P_n

Max indep. set

Min V. Cover

$\alpha(P_n) = \lceil \frac{n}{2} \rceil$

$\beta(P_n) = \lfloor \frac{n}{2} \rfloor$

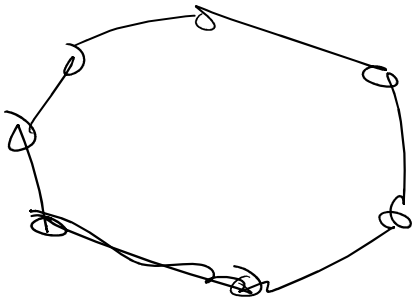
clique no.

Min dominating set

$\gamma(P_n) = \lceil \frac{n}{3} \rceil$

(v) Cycle graph on n vertices

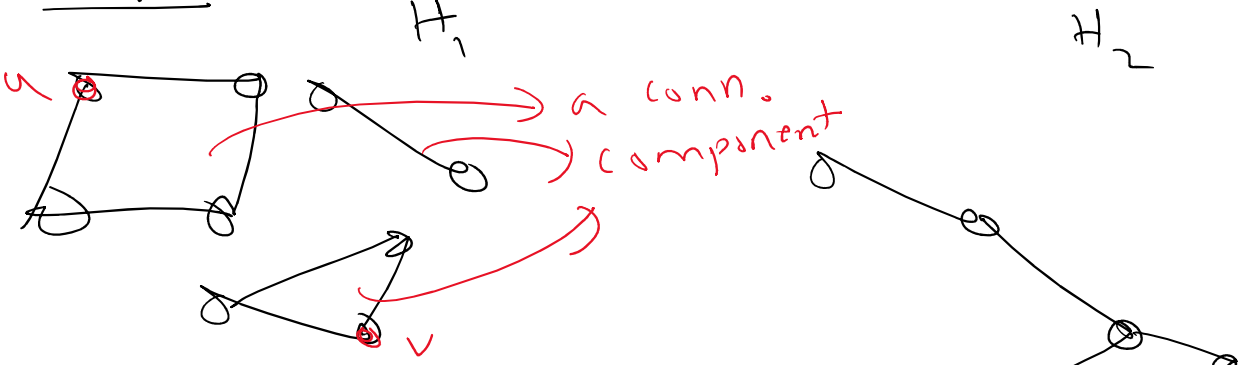
$w(P_n) = 2$



Connectivity

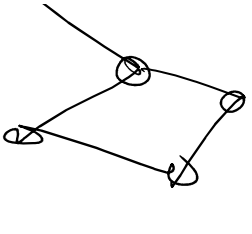
A graph G is connected if there is a path between every pair of vertices in G.

Example.





(i)



(ii)

H_1 is not connected.

H_2 is connected.

Defn.1 A graph G is k-vertex-connected (edge) if it cannot be disconnected by removing less than k vertices. (edges)

Convention: A single vertex graph is disconnected.

Defn 2: The edge vertex connectivity of a graph G, denoted $\kappa(G)$ / $\lambda(G)$, is the maximum k such that G is k-vertex (edge) connected.

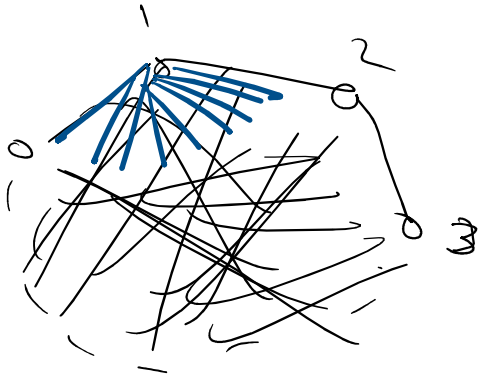
k-vertex connected $\Rightarrow (k-1)$-vertex conn.

Examples

minimum no. of vertices (edges) to be deleted to disconnect G.

Complete graph (K_n)

n



vertex connectivity

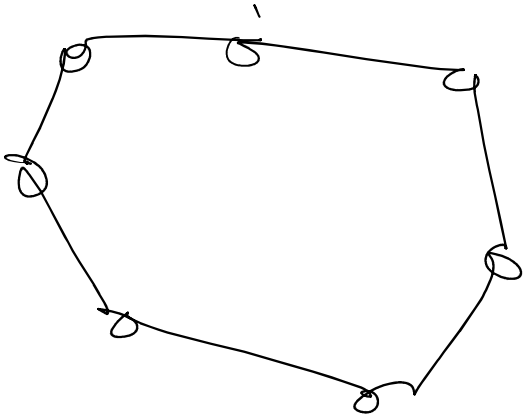
$\kappa(K_n) = n-1$

$\lambda(K_n) = n-1$



$\kappa(C_n) = 2$

$\lambda(C_n) = 2$



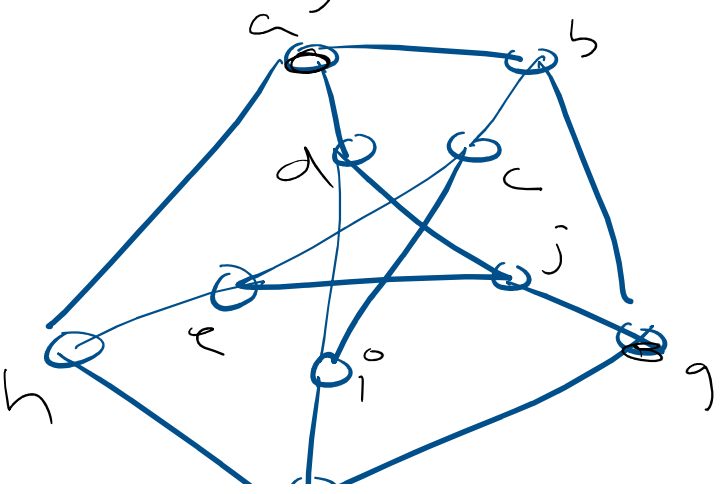
C_n

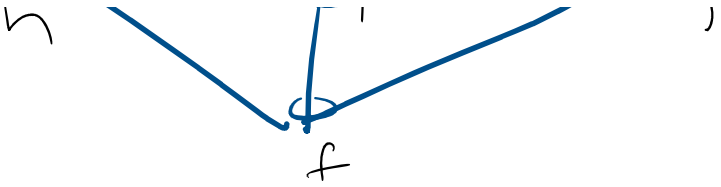


Prop: Let G be a connected graph on n vertices. Then, there exists an ordering of its vertices, say $v_1, v_2, \ldots, v_n$, such that $G[\{v_1, \ldots, v_i\}]$ is connected, for all $2 \leq i \leq n$.

Proof:

Notation: For any $S \subseteq V(G)$, we use $G[S]$ to denote the subgraph of G induced by S.





a, g, f, b, j - - -

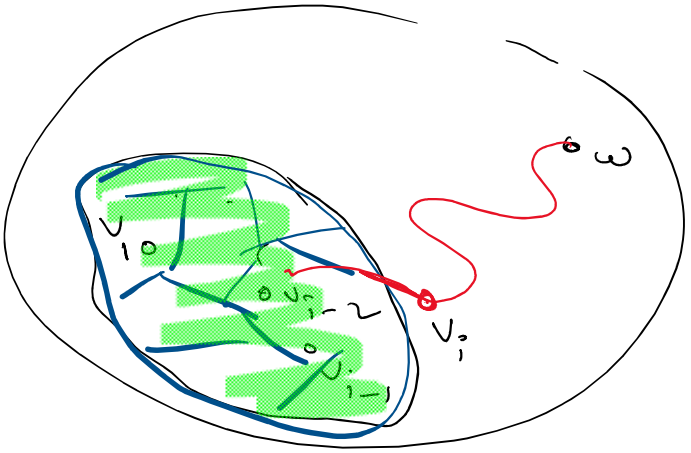
$\phi = (\quad)\wedge(\bar{x}_1 \vee \bar{x}_4 \vee x_5)\wedge(x_1 \vee x_2 \vee \bar{x}_3)\wedge(\quad)$ - -

n variables.

E.T.H. $=$ $2^{O(n)}$ time

$1-\delta^n = 2^{c_0 n}$, when c is a const.

Proof of prop:

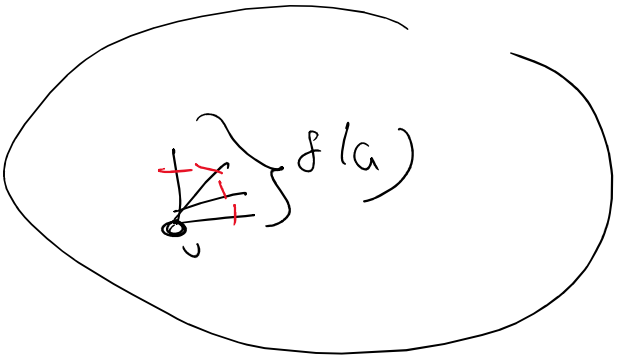


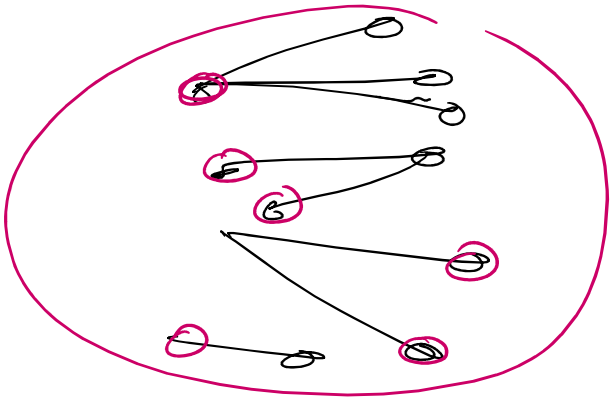
Prop: $\delta(G) \geq \chi(G) \geq K(G)$

min degree ① edge ② vertex

min degree ① edge conn. ② vertex conn.

Proof: Proving ①



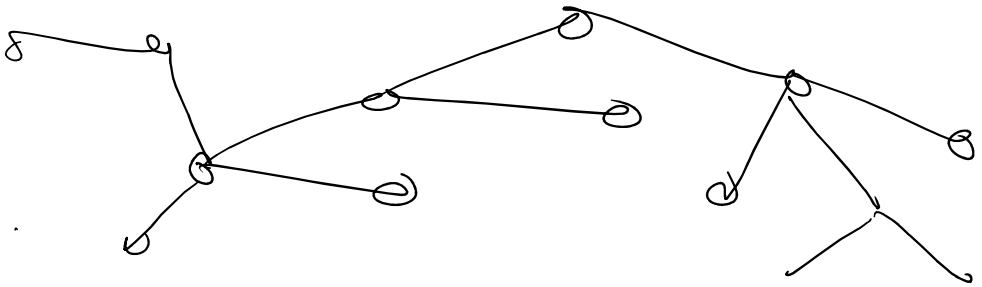


Trees.

An acyclic graph is called a FOREST.

A connected acyclic graph is called a TREE.

example.



Theorem: TFAE (The Following Are Equivalent)

(i) T is a tree,

(ii) Any two vertices of T are linked by a

(ii) Any two vertices of T are linked by a unique path in T;

(iii) T is minimally connected, i.e., $T-e$ is disconned for every edge e in T.

(iv) T is maximally acyclic, i.e. T containg no cycle but $T+xy$ contains a cycle, for every non-adjacent pair x, y.

Proof:

Understanding minimal/maximal vs minimum/maximum.

Example.



$S' = \{b\}$ is a minimum V.C

$T' = \{a, c\}$ is a maximum I.S.

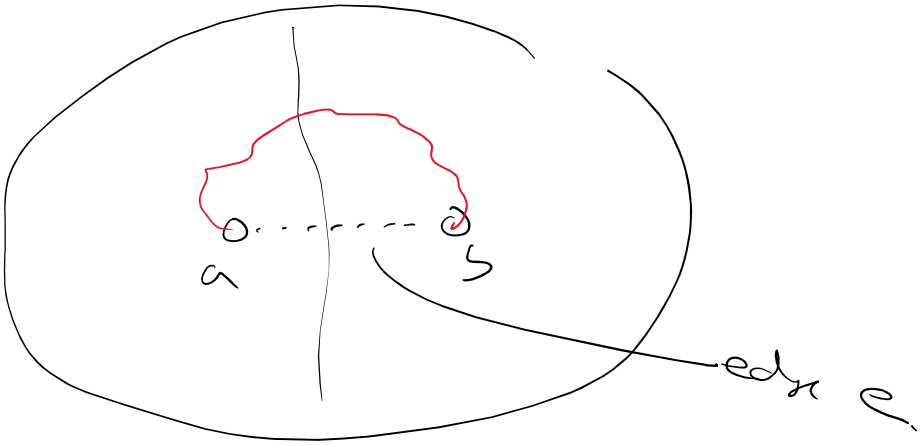
$S = \{a, c\}$ is a minimal vertex cover

and

$T = \{b\}$ is a maximal independent set

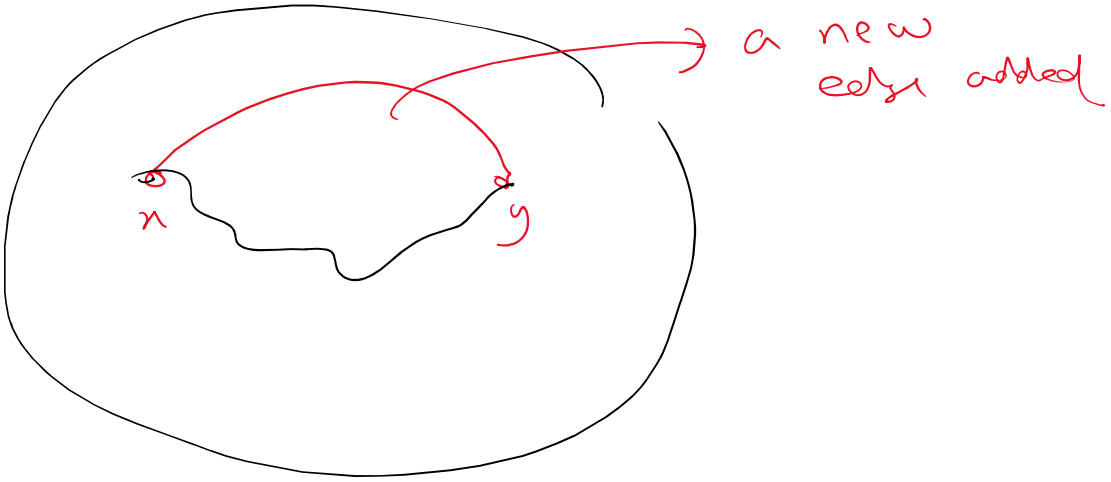
(ii) $\Rightarrow$ (iii)





(iii) $\Rightarrow$ (iv)

Suppose T is not maximally acyclic.



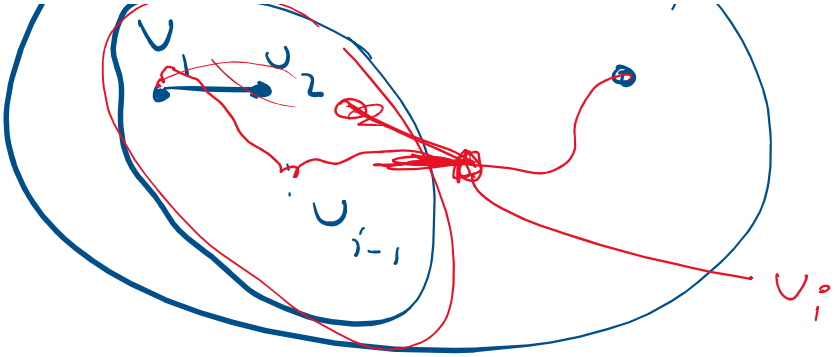
(iv) $\Rightarrow$ (i)

OR

maximally acyclic $\Rightarrow$ Connected

Corollary 1.5.2





Spanning tree

